

Curriculum Vitae

David Simons, BSc, MBBS, DTM&H, MSc, PhD

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Professional Appointments

Birgitta Sintring Independent Research Fellow	Sept 2026 – Present
<i>Department of Ecology and Genetics, Uppsala University, Sweden</i>	
External Expert	2025 – Present
<i>European Centre for Disease Prevention and Control (ECDC), Stockholm, Sweden</i>	
Collaborator, SCAPES project	2026 – Present
<i>The Pennsylvania State University, USA</i>	
Postdoctoral Scholar	2023 – 2026
<i>The Pennsylvania State University, Department of Anthropology & Center for Infectious Disease Dynamics, USA</i>	
Fellow-in-Residence	2023 – Present
<i>The VERENA Institute, USA</i>	
Senior Analyst, Outbreak Surveillance Team	2021 – 2022
<i>UK Health Security Agency, London, UK</i>	
Resident Doctor, Hospital for Tropical Diseases	2019 – 2022
<i>University College London Hospital, London, UK</i>	
Visiting Researcher	2020
<i>EcoHealth Alliance, New York, USA</i>	
Locum Senior House Officer	2018 – 2019
<i>Various NHS Hospitals, London, UK</i>	
Foundation Doctor	2016 – 2018
<i>North-East London Foundation Training School, London, UK</i>	
Research Assistant, Dept. of Microbiology, Tumour & Cell Biology	2012
<i>Karolinska Institute, Stockholm, Sweden</i>	
Clinical Research Scientist (Industrial Placement)	2008 – 2009
<i>GlaxoSmithKline, Harlow, UK</i>	

Education

PhD, Epidemiology of Emerging Zoonotic Infectious Diseases	2024
<i>University of London (RVC, UCL, LSHTM), London, UK</i>	
MSc, Tropical Medicine and International Health	2019
<i>London School of Hygiene and Tropical Medicine, London, UK</i>	
Professional Diploma in Tropical Medicine and Hygiene (DTM&H)	2018
<i>London School of Hygiene and Tropical Medicine, London, UK</i>	
Membership of the Royal College of Physicians (Part 1)	2017
<i>The Royal College of Physicians, London, UK</i>	
MBBS, Bachelor of Medicine and Bachelor of Surgery	2016
<i>King's College London, GKT School of Medical Education, London, UK</i>	
MSc & BSc, Natural Sciences	2011
<i>University of Leeds, Leeds, UK</i>	

Research Funding and Grants

Principal Investigator, Birgitta Sintring Fellowship (Funded)	2026 – Present
<i>Functional connectivity and microbial transmission pathways in Fennoscandian rodents. Uppsala University.</i>	
Principal Investigator, Vetenskapsrådet (VR) Starting Grant (Under Review)	2026
<i>HOST-V: Harnessing One-health Surveillance for Transforming Vaccine Trials for Epidemic Threats. SEK 6,000,000</i>	
Principal Investigator, Vetenskapsrådet (VR) Sweden-Korea Research Collaboration (Funded)	2026 – 2027
<i>An AI-Directed One Health Risk Prediction Framework for Active Learning in Rodent-Borne Zoonoses. Uppsala University, with Chungnam National University.</i>	
Co-Principal Investigator, FORMAS Project Grant, CASCADE (Under Review)	2026
<i>Climate change impacts on public health and zoonotic dynamics in Sweden.</i>	
Co-Principal Investigator, VERENA Institute Grant (Funded)	2023 – 2024
<i>Project ArHa: Arenaviruses and Hantaviruses. \$5,000</i>	
Principal Investigator, PhD Studentship (BBSRC LIDo DTP - Funded)	2019 – 2024
<i>Investigating rodent ecology to understand Lassa fever spillover. £85,000</i>	
Principal Investigator, EcoHealth Alliance Grant (Funded)	2020
<i>Forecasting Global Veterinary Disease: Modelling African Swine Fever. \$5,000</i>	
Marie Skłodowska-Curie Postdoctoral Fellowship (Awarded Seal of Excellence)	2025
<i>HOST-V framework score evaluated in top-tier threshold globally.</i>	

Publications

64. Imirzian N, Trebski A, Johnson E, Hewitson M, **Simons D**, Harden C, Friant S, Redding D (2026). ‘SiteTool’: a ‘Shiny’ application for field site selection and evaluation. *Ecography*.
63. **Simons D**, Watson-Jones D, Manno D, Friant S (2026). A One Health trial design to accelerate Lassa fever vaccines. *Nature Medicine*.
62. Catchen MD, Banville F, Boutin AC, Brookson CB, Carlson CJ, Dansereau G, ..., **Simons D**, et al. (2026). A protocol for biodiversity-informed wildlife disease surveillance. *PLOS Ecosystems*.
61. **Simons D**, Goyal R, Bangura U, Gibb R, Rushton B, Sondufu D, et al. (2026). Contact Networks of Small Mammals Highlight Potential Transmission Foci of Mammarenavirus lassaense. *The American Journal of Tropical Medicine and Hygiene*.
60. **Simons D**, Harden C, Imirzian N, Thompson KET, Ifebueme NM, Eziechina S, et al. (2026). Local heterogeneity in Lassa fever serology in rural Nigeria: Implications for vaccine trial site selection. *PLOS Neglected Tropical Diseases*.
59. Harden C, Eziechina S, Ifebueme NM, **Simons D**, Moses L, Redding DW, Friant S (2026). Local Knowledge of Land Use Effects on Human-Rodent Interfaces in Lassa-Endemic Nigeria. *EcoHealth*.
58. **Simons D** (2026). Map Liberator: An open-source tool for recovering spatial epidemiological data from static situation reports. *medRxiv*.
57. Rivero R, **Simons D**, Damodaran L, Karegi I, Gurev S, Becker DJ, et al. (2026). Molecular and ecological determinants of effective reassortment in orthohantaviruses. *bioRxiv*.
56. Leppin C, Brown J, Garnett C, Kale D, Okpako T, **Simons D**, Perski O (2026). Optimising supervised machine learning algorithms predicting cigarette cravings and lapses for a smoking cessation just-in-time adaptive intervention (JITAI). *PLOS One*.
55. Friant S, **Simons D**, Harden C, Imirzian N, Bjornstad O, Gibb R, et al. (2026). Scaling Lassa Virus Dynamics within Anthropogenic Ecosystems (SCAPES) Study Protocol: a mixed-methods observational cohort study of humans, rodents, and landscapes in Nigeria. *Wellcome Open Research*.
54. **Simons D** (2026). The socio-economic shield limits Lassa virus spillover in urban West Africa. *Epidemiology and Infection*.
53. **Simons D**, Rivero R, Rickard G, Martinez-Checa A, Gordon H, Redding DW, Seifert SN (2026). Viral reservoir status in small mammals emerges as a predictable life-history trait after correcting for surveillance bias. *bioRxiv*.
52. Schwantes CJ, Sánchez CA, Stevens T, Zimmerman R, Albery G, Becker DJ, ..., **Simons D**, et al. (2025). A minimum data standard for wildlife disease research and surveillance. *Scientific Data*.
51. **Simons D**, Gibb R, Bangura U, Sondufu D, Lamin J, Koninga J, et al. (2025). Land use gradients drive spatial variation in Lassa fever host communities in the Eastern Province of Sierra Leone. *Journal of Animal Ecology*.
50. **Simons D**, Rivero R, Guiote AMC, Gordon HLM, Milne GC, Rickard G, Redding DW, Seifert SN (2025). Protocol to produce a systematic Arenavirus and Hantavirus host-pathogen database: Project ArHa. *Wellcome Open Research*.
49. Friant S, Mistrick J, Luis AD, Harden C, **Simons D**, Fichet-Calvet E, et al. (2025). Reducing the threats of rodent-borne zoonoses requires an understanding and leveraging of three key pillars: disease ecology, synanthropy, and rodentation. *The Lancet Planetary Health*.

48. Streicker DG, Griffiths ME, Antia R, Bergner L, Bowman P, de Moraes MVdS, ..., **Simons D**, et al. (2024). Developing transmissible vaccines for animal infections. *Science*.
47. Perski O, Kale D, Leppin C, Okpako T, **Simons D**, Goldstein SP, Hekler E, Brown J (2024). Supervised machine learning to predict smoking lapses from Ecological Momentary Assessments and sensor data: Implications for just-in-time adaptive intervention development. *PLOS Digital Health*.
46. **Simons D**, Attfield LA, Jones KE, Watson-Jones D, Kock R (2023). A dataset of small-mammal detections in West Africa and their associated micro-organisms. *Gigabyte*.
45. Perski O, Li K, Pontikos N, **Simons D**, Goldstein SP, Naughton F, Brown J (2023). Classification of Lapses in Smokers Attempting to Stop: A Supervised Machine Learning Approach Using Data From a Popular Smoking Cessation Smartphone App. *Nicotine and Tobacco Research*.
44. Allen H, Tessier E, Turner C, Anderson C, Blomquist P, **Simons D**, et al. (2023). Comparative transmission of SARS-CoV-2 Omicron (B.1.1.529) and Delta (B.1.617.2) variants and the impact of vaccination: national cohort study, England. *Epidemiology and Infection*.
43. Arruda LB, Free HB, **Simons D**, Ansumana R, Elton L, Haider N, et al. (2023). Current sampling and sequencing biases of Lassa mammarenavirus limit inference from phylogeography and molecular epidemiology in Lassa fever endemic regions. *PLOS Global Public Health*.
42. **Simons D**, Attfield LA, Jones KE, Watson-Jones D, Kock R (2023). Rodent trapping studies as an overlooked information source for understanding endemic and novel zoonotic spillover. *PLOS Neglected Tropical Diseases*.
41. **Simons D**, Perski O, Shahab L, Brown J, Bailey R (2022). Association of smoking status with hospitalisation for COVID-19 compared with other respiratory viruses a year previous: a case-control study at a single UK National Health Service trust. *F1000Research*.
40. Vivancos R, Anderson C, Blomquist P, Balasegaram S, Bell A, Bishop L, ..., **Simons D**, et al. (2022). Community transmission of monkeypox in the United Kingdom, April to May 2022. *Eurosurveillance*.
39. Andrews N, Stowe J, Kirsebom F, Toffa S, Rieckard T, Gallagher E, ..., **Simons D**, et al. (2022). Covid-19 Vaccine Effectiveness against the Omicron (B.1.1.529) Variant. *New England Journal of Medicine*.
38. Kirsebom FCM, Andrews N, Stowe J, Toffa S, Sachdeva R, Gallagher E, et al. (2022). COVID-19 vaccine effectiveness against the omicron (BA.2) variant in England. *The Lancet Infectious Diseases*.
37. Blomquist PB, Bridgen J, Bray N, O'Connell AM, West D, Groves N, ..., **Simons D**, et al. (2022). Enhancing epidemiological surveillance of the emergence of the SARS-CoV-2 Omicron variant using spike gene target failure data, England, 15 November to 31 December 2021. *Eurosurveillance*.
36. Haider N, Guitian J, **Simons D**, Asogun D, Ansumana R, Honeyborne I, et al. (2022). Increased outbreaks of monkeypox highlight gaps in actual disease burden in Sub-Saharan Africa and in animal reservoirs. *International Journal of Infectious Diseases*.
35. **Simons D** (2022). Lassa fever cases suffer from severe underreporting based on reported fatalities. *International Health*.
34. Barnard RC, Davies NG, Centre for Mathematical Modelling of Infectious Diseases COVID-19 working group, Munday JD, Lowe R, Knight GM, ..., **Simons D**, et al. (2022). Modelling the medium-term dynamics of SARS-CoV-2 transmission in England in the Omicron era. *Nature Communications*.
33. Nightingale ES, Abbott S, Russell TW, CMMID Covid-19 Working Group, Rees EM, Eggo RM, ..., **Simons D**, et al. (2022). The local burden of disease during the first wave of the COVID-19 epidemic in England: estimation using different data sources from changing surveillance practices. *BMC Public Health*.

32. Leclerc QJ, Nightingale ES, Abbott S, CMMID COVID-19 Working Group, Rees EM, Eggo RM, ..., **Simons D**, et al. (2021). Analysis of temporal trends in potential COVID-19 cases reported through NHS Pathways England. *Scientific Reports*.
31. Osman AY, Elmi SA, **Simons D**, Elton L, Haider N, Khan MA, et al. (2021). Antimicrobial Resistance Patterns and Risk Factors Associated with Salmonella spp. Isolates from Poultry Farms in the East Coast of Peninsular Malaysia: A Cross-Sectional Study. *Pathogens*.
30. Davis EL, Lucas TCD, Borlase A, Pollington TM, Abbott S, Ayabina D, ..., **Simons D**, et al. (2021). Contact tracing is an imperfect tool for controlling COVID-19 transmission and relies on population adherence. *Nature Communications*.
29. The SAFER Investigators and Field Study Team, Hellewell J, The Crick COVID-19 Consortium, CMMID COVID-19 working group, Russell TW, Beale R, et al. (2021). Estimating the effectiveness of routine asymptomatic PCR testing at different frequencies for the detection of SARS-CoV-2 infections. *BMC Medicine*.
28. Blunsum AE, Perkins JS, Arshad A, Bajpai S, Barclay-Elliott K, Brito-Mutunayagam S, ..., **Simons D**, et al. (2021). Evaluation of the Implementation of the 4C Mortality Score in United Kingdom hospitals during the second pandemic wave. *medRxiv*.
27. Elmi SA, **Simons D**, Elton L, Haider N, Abdel Hamid MM, Shuaib YA, et al. (2021). Identification of Risk Factors Associated with Resistant Escherichia coli Isolates from Poultry Farms in the East Coast of Peninsular Malaysia: A Cross Sectional Study. *Antibiotics*.
26. Munday JD, Sherratt K, Meakin S, Endo A, Pearson CAB, Hellewell J, ..., **Simons D**, et al. (2021). Implications of the school-household network structure on SARS-CoV-2 transmission under school reopening strategies in England. *Nature Communications*.
25. Leclerc QJ, Fuller NM, Keogh RH, Diaz-Ordaz K, Sekula R, Semple MG, ..., **Simons D**, et al. (2021). Importance of patient bed pathways and length of stay differences in predicting COVID-19 hospital bed occupancy in England. *BMC Health Services Research*.
24. Prem K, Zandvoort Kv, Klepac P, Eggo RM, Davies NG, Centre for the Mathematical Modelling of Infectious Diseases COVID-19 Working Group, Cook AR, Jit M (2021). Projecting contact matrices in 177 geographical regions: An update and comparison with empirical data for the COVID-19 era. *PLOS Computational Biology*.
23. Quilty BJ, Clifford S, Hellewell J, Russell TW, Kucharski AJ, Flasche S, ..., **Simons D**, et al. (2021). Quarantine and testing strategies in contact tracing for SARS-CoV-2: a modelling study. *The Lancet Public Health*.
22. Perski O, **Simons D**, Shahab L, Brown J (2021). Smoking, Nicotine, and COVID-19: Triangulation of Methods and Preregistration Are Required for Robust Causal Inference. *Nicotine & Tobacco Research*.
21. Clifford S, Quilty BJ, Russell TW, Liu Y, Chan YWD, Pearson CAB, et al. (2021). Strategies to reduce the risk of SARS-CoV-2 importation from international travellers: modelling estimations for the United Kingdom, July 2020. *Eurosurveillance*.
20. CMMID COVID-19 working group, Jarvis CI, Gimma A, van Zandvoort K, Wong KLM, Edmunds WJ (2021). The impact of local and national restrictions in response to COVID-19 on social contacts in England: a longitudinal natural experiment. *BMC Medicine*.
19. CMMID COVID-19 Working Group, Liu Y, Morgenstern C, Kelly J, Lowe R, Jit M (2021). The impact of non-pharmaceutical interventions on SARS-CoV-2 transmission across 130 countries and territories. *BMC Medicine*.
18. Nightingale ES, Brady OJ, CMMID Covid-19 working group, Yakob L (2021). The importance of saturating density dependence for population-level predictions of SARS-CoV-2 resurgence compared with density-independent or linearly density-dependent models, England, 23 March to 31 July 2020. *Eurosurveillance*.

17. LSHTM CMMID COVID-19 Working Group, Mburu CN, Ojal J, Chebet R, Akech D, Karia B, et al. (2021). The importance of supplementary immunisation activities to prevent measles outbreaks during the COVID-19 pandemic in Kenya. *BMC Medicine*.
16. Arruda LB, Haider N, Olayemi A, **Simons D**, Ehichioya D, Yinka-Ogunleye A, et al. (2021). The niche of One Health approaches in Lassa fever surveillance and control. *Annals of Clinical Microbiology and Antimicrobials*.
15. Sandmann FG, Davies NG, Vassall A, Edmunds WJ, Jit M, Sun FY, ..., **Simons D**, et al. (2021). The potential health and economic value of SARS-CoV-2 vaccination alongside physical distancing in the UK: a transmission model-based future scenario analysis and economic evaluation. *The Lancet Infectious Diseases*.
14. Gibbs H, Liu Y, Pearson CAB, Jarvis CI, Grundy C, Quilty BJ, ..., **Simons D**, et al. (2020). Changing travel patterns in China during the early stages of the COVID-19 pandemic. *Nature Communications*.
13. Chevance G, Hekler EB, Efoui-Hess M, Godino J, Golaszewski N, Gualtieri L, ..., **Simons D**, et al. (2020). Digital health at the age of the Anthropocene. *The Lancet Digital Health*.
12. Kucharski AJ, Klepac P, Conlan AJK, Kissler SM, Tang ML, Fry H, ..., **Simons D**, et al. (2020). Effectiveness of isolation, testing, contact tracing, and physical distancing on reducing transmission of SARS-CoV-2 in different settings: a mathematical modelling study. *The Lancet Infectious Diseases*.
11. Perski O, **Simons D**, West R, Michie S (2020). Face masks to prevent community transmission of viral respiratory infections: A rapid evidence review using Bayesian analysis.
10. Kapata N, Ihekweazu C, Ntoumi F, Raji T, Chanda-Kapata P, Mwaba P, ..., **Simons D**, et al. (2020). Is Africa prepared for tackling the COVID-19 (SARS-CoV-2) epidemic. Lessons from past outbreaks, ongoing pan-African public health efforts, and implications for the future. *International Journal of Infectious Diseases*.
9. Dexter D, **Simons D**, Kiyaga C, Kapata N, Ntoumi F, Kock R, Zumla A (2020). Mitigating the effect of the COVID-19 pandemic on sickle cell disease services in African countries. *The Lancet Haematology*.
8. Haider N, Yavlinsky A, **Simons D**, Osman AY, Ntoumi F, Zumla A, Kock R (2020). Passengers' destinations from China: low risk of Novel Coronavirus (2019-nCoV) transmission into Africa and South America. *Epidemiology and Infection*.
7. Russell TW, Golding N, Hellewell J, Abbott S, Wright L, Pearson CAB, ..., **Simons D**, et al. (2020). Reconstructing the early global dynamics of under-ascertained COVID-19 cases and infections. *BMC Medicine*.
6. **Simons D**, Brown J, Shahab L, Perski O (2020). Smoking and COVID-19: Rapid evidence review for the Royal College of Physicians, London (UK). *Qeios*.
5. **Simons D**, Shahab L, Brown J, Perski O (2020). The association of smoking status with SARS-CoV-2 infection, hospitalization and mortality from COVID-19: a living rapid evidence review with Bayesian meta-analyses (version 7). *Addiction*.
4. Emery JC, Russell TW, Liu Y, Hellewell J, Pearson CA, CMMID COVID-19 Working Group, ..., **Simons D**, et al. (2020). The contribution of asymptomatic SARS-CoV-2 infections to transmission on the Diamond Princess cruise ship. *eLife*.
3. Quaife M, van Zandvoort K, Gimma A, Shah K, McCreesh N, Prem K, ..., **Simons D**, et al. (2020). The impact of COVID-19 control measures on social contacts and transmission in Kenyan informal settlements. *BMC Medicine*.
2. Firth JA, Hellewell J, Klepac P, Kissler S, CMMID COVID-19 Working Group, Jit M, ..., **Simons D**, et al. (2020). Using a real-world network to model localized COVID-19 control strategies. *Nature Medicine*.
1. O'Neill BV, Bullmore ET, Miller S, McHugh S, **Simons D**, Dodds CM, et al. (2012). The relationship between fat mass, eating behaviour and obesity-related psychological traits in overweight and obese individuals. *Appetite*.

Invited Presentations

A selection of 36 invited talks and poster presentations at international conferences and institutions including the ECOWAS Lassa fever conference, ASTMH, EEID, LSHTM, and the University of Idaho. Full list available upon request.

Teaching, Mentoring, and Supervision

Thesis Supervision 2019 – Present
Joint or sole supervisor for 8 students on medical elective, undergraduate, and postgraduate research projects at UCL, LSHTM, and the Royal Veterinary College.

Graduate & Professional Lecturing 2021 – Present
Invited guest lecturer for postgraduate courses at LSHTM (DTM&H) and UCL (Infectious Diseases MSc).

Undergraduate Lecturing 2023 – Present
Lecturer for “Spillover: The Anthropology of Emerging Infectious Diseases” at The Pennsylvania State University.

Field & Laboratory Capacity Building 2019 – Present
Recruited, trained, and supervised multiple local field and laboratory teams in Sierra Leone and Nigeria in rodent sampling, molecular (PCR), and antibody-based (ELISA, LIPS) assays under BSL-3 conditions.

Professional Service

External Expert, ECDC 2025 – Present
Co-developing curriculum for an e-learning course on “Environmental Determinants of Communicable Diseases.”

Peer Review 26 peer reviews for 14 international journals; 2 book chapter reviews. Academic Editor at PLOS Neglected Tropical Diseases

Grant Review Reviewer for the Grant Agency of the Czech Republic.

Skills

Data Analysis & Programming

- **R:** Tidyverse, R Markdown/Quarto, Shiny, Stan, sf
- **GIS:** QGIS, ArcGIS
- **Version Control:** Git, GitHub
- **Databases:** SQL basics

Laboratory & Field

- Rodent trapping and handling
- Serological assays (ELISA)
- Molecular diagnostics (PCR)

Languages

- **English:** Native
- **Swedish:** In progress

Clinical

- Infectious Diseases
 - Tropical Medicine
 - Emergency Medicine
 - General Medical Council (GMC) Registered
-